Package 'rfcipDemand'

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Type Package

Title Estimate Federal Crop Insurance Program Demand Models

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Description Tools to construct county–crop–practice–plan–unit panels from the USDA RMA Summary of Business (SOBTPU) and related sources, and to estimate FCIP demand systems with two-way cluster-robust covariance. The pipeline standardizes coverage measures, merges price and instrument variables, adds rental-rate and price-index controls, reconciles county acreage (FSA/NASS), and produces diagnostics including robust first-stage F-tests. Methods align with the empirical design in ``The crop insurance demand response to premium subsidies Evidence from U.S. Agriculture" (Food Policy, 2023, 119(3)).

License GPL-3

URL https://github.com/ftsiboe/rfcipDemand

BugReports https://github.com/ftsiboe/rfcipDemand/issues

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Suggests knitr, rmarkdown, tibble, dplyr, purrr, readr, lavaan, piggyback, testthat (>= 3.0.0)

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Cite-us If you find it useful, please consider staring the repository and citing the following studies - Tsiboe, F., & Turner, D. (2023). ``The crop insurance demand response to premium subsidies - Evidence from U.S. Agriculture". Food Policy, 119(3). doi.org/10.1016/j.foodpol.2023.102505 - Tsiboe, F., & Turner, D. (2023). Econometric identification of crop insurance participation. Agricultural and Resource Economics Review. 52(3) 476-497. doi.org/10.1017/age.2023.13

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`adjust_agent_outcomes_by_elasticity`
Adjust FCIP outcomes by price elasticities

Description

Simulates changes in Federal Crop Insurance Program (FCIP) outcomes- coverage level, insured acres, liability, premium, subsidy, and indemnity- under an alternative premium-per-liability using elasticities of insured acres and/or coverage level.

Usage

```
adjust_agent_outcomes_by_elasticity(
  alternate_premium_per_liability,
  insured_acres_elasticity,
  coverage_level_elasticity,
  baseline_coverage_type,
  baseline_coverage_level,
  baseline_insured_acres,
  baseline_liability_per_acre,
  baseline_premium_per_liability,
  baseline_subsidy_per_premium,
  baseline_indemnity_per_acre,
  baseline_subsidy_percent,
  baseline_rate_differential,
  final_revenue_per_acre,
  assumption = 0,
  premium_subsidy_schedule = NULL,
  rate_differential_schedule = NULL
)
```

Arguments

`alternate_premium_per_liability`	Numeric. Alternative premium per dollar of liability.
`insured_acres_elasticity`	Numeric. Elasticity of insured acres w.r.t. price (percent basis).
`coverage_level_elasticity`	Numeric. Elasticity of coverage level w.r.t. price (percent basis).
`baseline_coverage_type`	Character. Baseline coverage type code (`"A"` or `"C"`).
`baseline_coverage_level`	Numeric in (0,1]. Baseline coverage level share.
`baseline_insured_acres`	Numeric. Baseline insured acres.
`baseline_liability_per_acre`	Numeric. Baseline liability per acre.
`baseline_premium_per_liability`	Numeric. Baseline premium per dollar of liability.
`baseline_subsidy_per_premium`	Numeric in (0,1). Subsidy share of total premium.
`baseline_indemnity_per_acre`	Numeric. Baseline indemnity per acre.
`baseline_subsidy_percent`	Numeric. Baseline subsidy percent.
`baseline_rate_differential`	Numeric. Baseline rate differential factor.
`final_revenue_per_acre`	Numeric. Revenue per acre (used for indemnity adjustment).
`assumption`	Integer (0,1,2,3). Scenario selector (see above). Default `0`.

`premium_subsidy_schedule`
: Optional numeric vector of length 8 corresponding to coverage levels `0.50`, `0.55`, `...`, `0.85`. Multiplicative factors applied to `baseline_subsidy_per_premium` at the scenario coverage. Defaults to 1's.

`rate_differential_schedule`
: Optional numeric vector of length 8 corresponding to coverage levels `0.50`, `0.55`, `...`, `0.85`. Multiplicative factors applied to `alternate_premium_per_liability` at the scenario coverage. Defaults to 1's.

Details

Assumptions (set via `assumption`):

- `0`: Fixed demand - both coverage and acres remain at baseline.
- `1`: Acres respond to price (elasticity gamma); coverage fixed.
- `2`: Coverage responds to price (elasticity tetha); acres fixed.
- `3`: Both acres and coverage respond.

Price response:

Price response is applied using the percent change in price:

$$\%\Delta p = 100 \times \left(\frac{\text{alt}}{\text{base}} - 1 \right),$$

and for any baseline quantity q with elasticity e_q:

$$q_{\text{new}} = q_{\text{base}} \left[1 + \left(e_q \times \frac{\%\Delta p}{100} \right) \right].$$

Coverage adjustment:

Coverage is rounded to the nearest 0.05 grid and truncated to the range $(0, 0.85)$. Values less than 0.50 are set to `0` (i.e., no coverage). Coverage type code is set to `"A"` for buy-up coverage and `"C"` for CAT coverage. For CAT policies, scenario coverage levels at 0.50/0.55 are adjusted to reflect CAT rules.

Per-acre liability and indemnity are updated using `adjust_indemnity_liability_per_acre`.

Optional schedules:

Coverage-specific scaling is possible via:

- `premium_subsidy_schedule`: length-8 numeric for coverage levels `0.50`, `0.55`, `...`, `0.85`. Multiplies the `baseline_subsidy_per_premium` at the scenario coverage.
- `rate_differential_schedule`: length-8 numeric for the same grid. Multiplies the `alternate_premium_per_li` at the scenario coverage.

Missing or zero inputs:

If either `insured_acres` or `coverage_level_percent` is `NA` or `0` after adjustment, both are reset to `0` and all dollar outcomes are set to `0`.

Elasticities are applied multiplicatively to baseline quantities. Coverage levels are snapped to the 0.05 grid and truncated. When schedules are supplied, subsidy shares and premium rates are rescaled relative to the baseline coverage before applying to the scenario coverage.

Value

A list with elements:

`coverage_type_code` Scenario coverage type (`"A"` or `"C"`).

`coverage_level_percent` Scenario coverage level share.

`insured_acres` Scenario insured acres.

`liability_amount` Total liability = acres * adjusted liability per acre.

`total_premium_amount` Total premium = liability * alt premium/liability.

`subsidy_amount` Subsidy = total premium * subsidy share.

`indemnity_amount` Total indemnity = acres * adjusted indemnity per acre.

`price_change_pct` Percent price change used for elasticity adjustments.

`adjust_indemnity_liability_per_acre` *Adjust liability and indemnity per acre at an alternative coverage level*

Description

Compute adjusted liability per acre and adjusted indemnity per acre when moving from a baseline coverage level to a new level. The method floors production-to-count at 0 and resolves five mutually exclusive cases via `data.table::fcase()`.

Usage

```
adjust_indemnity_liability_per_acre(
  coverage_level_percent,
  final_revenue_per_acre,
  baseline_coverage_level,
  baseline_liability_per_acre,
  baseline_indemnity_per_acre
)
```

Arguments

`coverage_level_percent`
: Numeric vector; alternative coverage level (proportion in (0,1)).

`final_revenue_per_acre`
: Numeric vector; revenue per acre at harvest (used when baseline had no indemnity and baseline coverage is lower).

`baseline_coverage_level`
: Numeric vector; baseline coverage level (proportion in (0,1)).

`baseline_liability_per_acre`
: Numeric vector; baseline liability per acre.

`baseline_indemnity_per_acre`
: Numeric vector; baseline indemnity per acre.

Details

- `production_to_count = pmax(baseline_liability_per_acre - baseline_indemnity_per_acre, 0)`
- `adj_Liability_per_acre = (baseline_liability_per_acre / baseline_coverage_level) * coverage_level_percent`
- Indemnity cases (I1-I5) selected with `data.table::fcase()`.

Value

A list with:

`adj_Liability_per_acre` Adjusted liability per acre.

`adj_Indemnity_per_acre` Adjusted indemnity per acre.

---

agCensusAcres *agCensusAcres*

Description

A combined dataset for agCensusAcres

Usage

```
data(agCensusAcres)
```

Format

A data frame with 15369 rows and 15 columns covering NA-NA.

Source

USDA NASS Quick Stats

---

agCensusBFR *agCensusBFR*

Description

A combined dataset for agCensusBFR

Usage

```
data(agCensusBFR)
```

Format

A data frame with 255 rows and 16 columns covering NA-NA.

Source

USDA NASS Quick Stats

`calibrate_fcip_demand_elasticities` *Calibrate FCIP demand elasticities*

Description

Fits a 2-equation FCIP demand system over an estimation window ending in `calibration_year` and returns capped elasticities (-2, 0) by disaggregation level using `fcip_demand_sys_estimate()`

Usage

```
calibrate_fcip_demand_elasticities(
  calibration_year,
  estimation_window,
  data,
  drawn_pools = NULL,
  disaggregate = NULL,
  control = rfcipDemand_controls()
)
```

Arguments

`calibration_year`	Integer. Last year of the estimation window.
`estimation_window`	Integer (>= 1). Number of years ending at `calibration_year`.
`data`	`data.table`. Input panel. Must include: `commodity_year`, `commodity_code`, `drawID`, `pool`, `insurance_plan_code`, `price`, `net_reporting_level_amount`, `coverage_level_percent_aggregate`, `rent`, `county_acreage`, `total_premium_amount`, `subsidy_amount`, `liability_amount`, `tau`, `subsidy_rate_65`, `subsidy_rate_75`.
`drawn_pools`	Optional `data.table` used to filter `data` via an inner join on intersecting columns.
`disaggregate`	Optional character vector of grouping variables passed to the estimator (e.g., `"commodity_code"` or `c("state","commodity_code")`).
`control`	a list of control parameters. see `rfcipDemand_controls`.

Details

Internally, the function (i) restricts `data` to the estimation years, (ii) collapses insurance plan codes and sets `price := 1` for non-RP crops, (iii) constructs log variables, trend and year dummies, and (iv) estimates the system with `rate` endogenous and `tau0` as the excluded instrument. Estimates are truncated to (-2, 0) and returned in wide form.

Value

`data.table` with columns: `disag`, `level`, `gamma_elasticity`, `theta_elasticity`.

`estimate_fcip_instruments` *Estimate FCIP Instrumental Variables (Unloaded Rates)*

Description

Uses historical FCIP rate data to build instrumented unloaded-rate variables following:

1. Tsiboe & Turner (2023), Econometric identification of crop insurance participation *Agricultural and Resource Economics Review*, 52(3):476-497. https://doi.org/10.1017/age.2023.13

Usage

```
estimate_fcip_instruments(year, statplan)
```

Arguments

`year`	Integer. The target crop year for which to construct instruments.
`statplan`	A data.table containing FCIP rate elements, including at least: **commodity_year** Year of the rate observation. **state_code, county_code** County identifiers. **commodity_code** Crop identifier. **insured_area, lcr, contiguous_state_code, contiguous_county_code** Fields required by `estimate_fcip_unloaded_rate()`.

Details

1. **Task list**: Identify all unique (state, county) pairs with data in the 2-21 years before `year`.
2. **Unloaded-rate calculation**: For each county in `task_list`, call `estimate_fcip_unloaded_rate()` on the same 2-21 year window to get `tau`. Errors return `NULL` so processing continues.
3. **Contiguous-county smoothing**:
   - Build a lookup table of contiguous counties (using `contiguous_county`).
   - For each contiguous group, compute the mean `tau` to get `tau_c`.
4. **Merge & fill**: Left-join the raw `adm` and `contiguous_adm`; replace any zero/NA/Inf `tau` with the group mean `tau_c` into `tau_sob`.
5. **Cleanup**: Drop helper columns (`tau`, `tau_c`), remove invalid rows, add `commodity_year`, and return the result.

Value

A data.table with one row per county-crop for the specified `year`, containing:

state_code, county_code, commodity_code Keys.

tau_sob Smoothed unloaded rate (uses contiguous-county means to fill zeros/NAs).

commodity_year The input `year`, repeated.

See Also

Other FCIP instruments: `get_yu2018_instrument()`

fcip_contiguous_county

fcip_contiguous_county

Description

A combined dataset for fcip_contiguous_county

Usage

```
data(fcip_contiguous_county)
```

Format

A data frame with 24307 rows and 12 columns covering NA-NA.

Source

USDA-RMA, Actuarial Data Master - A0123

fcip_demand_data_dispatcher

Build dataset to estimate Federal Crop Insurance Program (FCIP) demand (modular pipeline)

Description

End-to-end pipeline that: (1) prepares SOBTPU and coverage aggregates, (2) adds prices/instruments/rental rates/price index, (3) reconciles county acreage (FSA + NASS), and (4) finalizes bins/labels/pooling for demand estimation.

Usage

```
fcip_demand_data_dispatcher(
  study_years = 2001:(as.numeric(format(Sys.Date(), "%Y")) - 1),
  identifiers = c("commodity_year", FCIP_INSURANCE_POOL, "insurance_plan_code",
    "unit_structure_code")
)
```

Arguments

study_years	Integer vector of commodity years to include. Defaults to 2001:(as.numeric(format(Sys.Date(), "%Y")) - 1).
identifiers	Character vector of grouping keys that define the aggregation grain. Must be columns present in SOBTPU (e.g., "commodity_year", FCIP_INSURANCE_POOL, "insurance_plan_code", "unit_structure_code", and-if desired-additional keys like "commodity_code" or "practice_code"). Enrichment joins for recodes are performed only when the required keys are included in identifiers

Details

Aligned with Asche, Bekkerman, & Li (2023), *Food Policy*, 119(3):102505 (doi:10.1016/j.foodpol.2023.102505). Requires internet access to download release `.rds` assets and several in-memory lookup tables (see stage docs).

Value

A `data.table` ready for FCIP demand estimation.

`fcip_demand_elasticities_lavaan` *Calibrate FCIP demand elasticities via IV-SEM (lavaan)*

Description

Fits a just-identified IV-style SEM where `instr_rate` instruments `tilda_rate`, and `tilda_rate` enters two outcome equations (`tilda_ghamma`, `tilda_theta`). Endogeneity is encoded via residual correlations between `tilda_rate` and each outcome. The instrument is excluded from the outcome disturbances.

Usage

```
fcip_demand_elasticities_lavaan(
  data,
  estimator = c("ML", "MLR"),
  missing = c("fiml", "listwise"),
  control = rfcipDemand_controls()
)
```

Arguments

`data`	data.frame with cols: instr_rate, tilda_rate, tilda_gamma, tilda_theta
`estimator`	"ML" or "MLR" (default "MLR" = robust SE/tests)
`missing`	"fiml" or "listwise" (default "fiml")
`control`	a list of control parameters. see `rfcipDemand_controls`.

Details

Constraints imposed:

1. b1 < 0, 2) b2 < 0, 3) b1 + b2 + b1*b2 < 0

Value

A data.table of parameter estimates and logical convergence flag

fcip_demand_sys_coeff_table

Tidy coefficient table with cluster-robust SEs (from supplied VCOV)

Description

Builds a clean coefficient table for a `systemfit` model using a **user-supplied covariance matrix** (e.g., two-way clustered from `fcip_demand_sys_vcov()`). Estimates come from `coef(fit)`, standard errors from `diag(vcMat)`, then Z-scores and two-sided normal p-values are computed. The `demand` column is inferred from the equation prefix in the coefficient names:

- `"gamma_*"` to `"gamma"`
- `"theta_*"` to `"theta"` Otherwise the prefix itself is used.

Usage

```
fcip_demand_sys_coeff_table(fit, vcMat, p_digits = 5)
```

Arguments

`fit`	A fitted `systemfit` object.
`vcMat`	A covariance matrix conformable with `coef(fit)`. Row/column names are used to align; if missing, positional alignment is assumed.
`p_digits`	Integer; number of digits to keep for p-values (default `5`).

Value

A `data.frame` with columns: `demand`, `coef`, `Estimate`, `StdError`, `Zvalue`, `Pvalue`.

fcip_demand_sys_effect

Delta-method "total protection response"

Description

Combines equation-specific effects into a single "total" effect for each regressor in `c(fields$endogenous, fields$included)` using `car::deltaMethod` and a supplied covariance matrix.

Usage

```
fcip_demand_sys_effect(fit, vcMat, fields, data)
```

Arguments

`fit`	A fitted `systemfit` object (the structural system).
`vcMat`	Covariance matrix conformable with `coef(fit)` (e.g., from `fcip_demand_sys_vcov()`).
`fields`	Named list carrying model fields; must include `outcome`, `endogenous`, and `included`.
`data`	Estimation data used to check variable availability and build delta-method expressions.

Value

A data.frame with rows demand="total" and columns: demand, coef, Estimate, StdError, Zvalue, Pvalue.

`fcip_demand_sys_estimate` *System estimator (modular wrapper; preserves original outputs)*

Description

Estimates a two-equation system with an endogenous regressor across disaggregation levels. The pipeline is: (1) per-level sample selection (min n per commodity_year), (2) partialling-out / tilda transforms, (3) system estimation (e.g., IV/3SLS via internal helpers), (4) clustered variance estimation, (5) delta-method totals, (6) optional constrained re-estimation of elasticities (lavaan), (7) diagnostics, and (8) row-binding results across levels.

Usage

```
fcip_demand_sys_estimate(
  model,
  data,
  constrained_elasticities = FALSE,
  control = rfcipDemand_controls()
)
```

Arguments

model	List specifying the system; required elements are: • outcome (character(2)): names of the two outcomes in data. • endogenous (character(1)): endogenous regressor name. • included (character): included (exogenous) regressors. • disag (character(1) or NULL): disaggregation key column in data. • FE (logical): include fixed effects in the internal run (handled by helpers). Optional elements: • excluded (character or NULL): excluded instruments. • partial (character or NULL): variables to partial out (tilda). • restrict (logical): pass-through to internal restricted estimation. • name (character(1)): label carried to the output.
data	data.frame/data.table containing all referenced columns in model plus pool and commodity_year. Columns in model$outcome, model$endogenous, model$included, and (if used) model$excluded and model$partial must exist in data.
constrained_elasticities	Logical (default FALSE). If TRUE, re-estimate the elasticities via a constrained SEM (lavaan) and, where applicable, replace positive elasticity estimates with constrained ones. See **Constrained elasticities (optional)**.
control	a list of control parameters. see rfcipDemand_controls.

Details

Inputs and preprocessing

- `model$outcome` must be a character vector of length 2 giving the two outcome column names in `data`. Internally these are mapped to `gamma` and `theta` for estimation convenience.
- `model$disag` is the name of the disaggregation key. If `NULL`, a dummy `"full_sample"` key is created.
- The disaggregation key is coerced to `character`.
- For each level of `model$disag`, the function keeps only levels that have **at least 30 observations per** `commodity_year`. Levels failing this threshold are dropped.

Per-level estimation For each retained level, the function calls internal helpers (`fcip_demand_sys_run`, etc.) to (i) partially out controls if requested, and (ii) estimate the system with clustered VCOV and delta-method totals. Errors at the level are caught and the level is skipped (no hard stop).

Constrained elasticities (optional) When `constrained = TRUE`, elasticities on the endogenous regressor are re-estimated per level via a lavaan SEM with sign restrictions (internal helper `fcip_demand_elasticities_l` using `estimator = "MLR"`, `missing = "listwise"`). The constrained estimates replace any positive system estimates for the elasticities; a logical flag `constrained` marks levels where a replacement occurred. The returned columns are reduced to `disag`, `level`, `demand`, `constrained`, and `Estimate`.

Returned shape

- If `constrained_elasticities = FALSE` (default), returns the full per-level system output from `fcip_demand_sys_run` with additional columns: `disag`, `level`, and rounded `Zvalue`, `Pvalue`.
- If `constrained = TRUE`, returns a compact table with `disag`, `level`, `demand`, `constrained`, `Estimate` after merging the constrained elasticities.

Value

A `data.table` aggregating results across all disaggregation levels. The column set depends on `constrained` (see **Returned shape**).

`fcip_demand_sys_fit` *Build systemfit formulas and estimate the system*

Description

Constructs the list of structural equations (`g`) and instrument sets (`h`), then runs `systemfit()` using OLS (when no excluded instruments) or 3SLS-GMM (when excluded instruments are present).

Usage

```
fcip_demand_sys_fit(
  data,
  fields,
  tilda_included,
  tilda_endogenous,
  tilda_excluded
)
```

Arguments

data	Estimation data.frame/data.table containing the tilda_* and instr_* variables referenced by the formulas.
fields	Named list with at least outcome, included, endogenous, and optionally excluded.
tilda_included	Character vector of residualized included regressor names (e.g., "tilda_x1").
tilda_endogenous	Character vector of residualized endogenous regressor names (e.g., "tilda_z1").
tilda_excluded	Character vector of instrument names (e.g., "instr_z1"), or NULL when no excluded instruments are used.

Value

A list with elements:

fit	Fitted systemfit object.
g	List of structural formulas.
h	List of instrument formulas.

---

fcip_demand_sys_partial

Residualize ("partial out") and build tilded / instrument variables

---

Description

If excluded instruments exist, runs first-stage OLS for each endogenous variable e: e ~ 1 + partial + included + excluded, storing the fitted values as instr_e. If partial is non-empty, it then regresses instr_e ~ 1 + partial and replaces instr_e <- instr_e - fitted(instr_e ~ partial) (i.e., removes the partial component; conceptually $\widehat{\text{instr}}_e(\text{partial})$).

Outcomes, included, and endogenous variables are residualized on partial to create tilda_<var> (or copied if partial is empty).

Usage

```
fcip_demand_sys_partial(data, fields, partial_override = NULL)
```

Arguments

data	A data.frame/data.table with referenced variables.
fields	List with: outcome, endogenous, included, optional excluded, optional partial.
partial_override	Optional character vector to override fields$partial.

Details

Uses defensive checks so absent columns are ignored (with a warning) rather than erroring. If partial is empty, residualization is a no-op and tilda_* simply copy the originals. Formulas are constructed via stats::reformulate() to avoid paste/quoting pitfalls.

Value

List with data, tilda_included, tilda_endogenous, tilda_excluded.

fcip_demand_sys_prep *Prepare data for analysis*

Description

Drops incomplete/invalid rows, removes constant partials, and optionally demeans via a fixed-effects helper.

Usage

```
fcip_demand_sys_prep(data, fields)
```

Arguments

data	Estimation dataset that already contains all columns referenced by fields.
fields	Named list: outcome, endogenous, included, excluded (opt), partial (opt), FE (logical), disag (column name).

Value

A list: data (prepped), NFE (number of FE), partial (possibly reduced).

fcip_demand_sys_run *Orchestrate analysis*

Description

Runs the full pipeline: prep -> partial/tilda creation -> systemfit -> two-way clustered VCOV -> delta-method totals -> optional restricted step -> diagnostics; then returns a tidy coefficient table with metadata.

Usage

```
fcip_demand_sys_run(data, fields)
```

Arguments

data	Estimation dataset
fields	Named list carrying model fields (see fcip_demand_estimation()), including disag, FE, outcome, endogenous, included, optional excluded, partial, restrict, and name.

Value

A data.frame with columns demand, coef, Estimate, StdError, Zvalue, Pvalue and meta-columns model, endogenous, FE, name, disag

fcip_demand_sys_tests *System diagnostics: two-way robust first-stage F (+ optional approx. J)*

Description

Produces diagnostics **without** re-running GMM:

- **FTest**: joint relevance of excluded instruments in each first stage, using the same two-way (pool by crop year) cluster-robust covariance via `fcip_demand_sys_vcov()` with `kind = "lm"`. Reports the **minimum** F across endogenous regressors.
- **JTest** (optional): an *approximate* over-identification test computed as the sum of per-equation Sargan statistics $J_k \approx n_k R_k^2$ from regressions of equation residuals on that equation's instrument set. This is a quick check (not the system Hansen J).

Usage

```
fcip_demand_sys_tests(g, h, data, fit, NFE, approx_j = FALSE)
```

Arguments

g	List of system equations (the same formulas passed to `systemfit`).
h	List of instrument formulas (the same formulas passed to `systemfit`).
data	Estimation `data.frame`/`data.table` containing all variables in `g`/`h` plus clustering columns `pool` and `crop_yr`.
fit	A fitted `systemfit` object (used for `N` and `residCov_*` extraction).
NFE	Integer: number of absorbed fixed effects (for reporting only).
approx_j	Logical, compute the approximate (non-robust) Sargan J as described above. Default `FALSE` (returns `NA` for `JTest`).

Value

A `data.frame` with rows: `N`, `NFE`, `residCov_11`, `residCov_22`, `residCov_12`, `JTest`, `FTest`.

fcip_demand_sys_vcov *Two-way cluster-robust covariance for FCIP demand models*

Description

Computes a Cameron-Gelbach-Miller two-way cluster-robust covariance matrix using inclusion-exclusion: $V = V_{pool} + V_{year} - V_{pool_year}$. Works for both `systemfit` (stacked system) and `lm` (first-stage).

Usage

```
fcip_demand_sys_vcov(
  object,
  data,
  kind = c("systemfit", "lm"),
  pool_col = "pool",
  year_col = "commodity_year",
  NFE = 0L,
  n_partial = 0L,
  n_eq = NULL
)
```

Arguments

object	Fitted model: either a systemfit or lm.
data	Estimation data containing pool and year identifiers.
kind	One of c("systemfit","lm"). If omitted, auto-detected.
pool_col	Name of the pool/cluster id column in data (default "pool").
year_col	Name of the year/time id column in data (default "crop_yr").
NFE	Integer; number of absorbed fixed effects (for df rescaling).
n_partial	Integer; count of variables partialed out per equation.
n_eq	Integer; number of equations (length(object$eq) for systemfit, 1 for lm). You can override if needed.

Details

Rescaling. Let n be the number of observations (stacked across equations for systemfit). With k_old the number of coefficients and $k_new = k_old + NFE + n_partial * n_eq$, the returned matrix is scaled by $(n - k_old - 1)/(n - k_new - 1)$.

Row alignment (lm). Rows used by lm are inferred from rownames(model.matrix(object)). If they cannot be mapped back to data, the first nobs(object) rows are used.

Value

Covariance matrix aligned with coef(object).

---

fcip_elasticity_limits

rfcipDemand Helper Datasets

Description

A combined dataset for fcip_elasticity_limits

Usage

```
data(fcip_elasticity_limits)
```

Format

A data frame with 1 rows and 4 columns covering NA-NA.

Source

Elasticity limits determined as 50% of the magnitude of the program level elasticities

---

FCIP_INSURANCE_POOL *Insurance pool identifier fields*

Description

A character vector of column names that together define a unique insurance pool in the Federal Crop Insurance Program (FCIP).

Usage

```
FCIP_INSURANCE_POOL
```

Format

A `character` vector of field names.

Details

Insurance pools represent the most granular level of rate making within FCIP. Each pool is uniquely identified by the combination of:

- **state_code**: State FIPS code
- **county_code**: County FIPS code
- **commodity_code**: Crop commodity code
- **type_code**: Crop type (e.g., grain vs. silage)
- **practice_code**: Production practice (e.g., irrigated, organic)

Value

A `character` vector specifying the columns used to define each FCIP insurance pool.

Examples

```
## Not run:
# Default insurance pool fields
FCIP_INSURANCE_POOL

# Override to a subset of the original fields
rFarmPolicySim:::FCIP_INSURANCE_POOL <- c(
  "state_code", "county_code", "commodity_code"
)

## End(Not run)
```

fcip_recodes_commodity_groupings
fcip_recodes_commodity_groupings

Description

A combined dataset for fcip_recodes_commodity_groupings

Usage

```
data(fcip_recodes_commodity_groupings)
```

Format

A data frame with 3729 rows and 10 columns covering 1997-2027.

Source

USDA-RMA, Actuarial Data Master - A00400 and A00420 supplemented data from legacy ADM files

fcip_recodes_insurance_plan
fcip_recodes_insurance_plan

Description

A combined dataset for fcip_recodes_insurance_plan

Usage

```
data(fcip_recodes_insurance_plan)
```

Format

A data frame with 826 rows and 10 columns covering 1989-2027.

Source

USDA-RMA, Actuarial Data Master - A00460 supplemented data from legacy ADM files

fcip_recodes_practice *fcip_recodes_practice*

Description

A combined dataset for fcip_recodes_practice

Usage

```
data(fcip_recodes_practice)
```

Format

A data frame with 30799 rows and 8 columns covering 1997-2027.

Source

USDA-RMA, Actuarial Data Master - A00510 supplemented data from legacy ADM files

fcip_recodes_type *fcip_recodes_type*

Description

A combined dataset for fcip_recodes_type

Usage

```
data(fcip_recodes_type)
```

Format

A data frame with 237671 rows and 7 columns covering 1999-2026.

Source

Generated internally, using harmonize_crop_type_codes()

`fixed_effect_model_data_prep` *Prepare and demean data for fixed-effects models*

Description

This function

1. Filters to complete cases on the specified panel, time, weight, variables, and output
2. If `output` is NULL, creates a dummy output column filled with 1s
3. Drops any panel with only one observation
4. Computes within-panel means for the output + each variable in `varlist` (`_mean_i`)
5. Computes overall sample means for the same set of variables (`_mean`)
6. Replaces each variable in `varlist` by `value - within_panel_mean + overall_mean`

Usage

```
fixed_effect_model_data_prep(
  data,
  varlist,
  panel,
  time,
  wvar = NULL,
  output = NULL
)
```

Arguments

`data`	A data.frame or data.table containing the data.
`varlist`	Character vector of variable names to be demeaned.
`panel`	Character vector of column name(s) defining the panel identifier.
`time`	Character scalar name of the time variable.
`wvar`	Character scalar name of a variable to keep but *not* demean (optional, default NULL).
`output`	Character scalar name of an output variable whose means are computed but not altered; if NULL, a dummy column named `"output"` is created (optional, default NULL).

Value

A list with components

- **data**: a data.table containing
  - the original `panel`, `time`, `wvar`, `varlist`, and `output` columns
  - two mean columns for each of `c(output, varlist)`: `<name>_mean_i` (within-panel) and `<name>_mean` (overall)
- **NFE**: the number of panels with more than one observation

format_fcip_demand_table *Table: Crop Insurance Demand System for US Federal Crop Insurance Pools (2001/22)*

Description

Build a two-column, GitHub-safe panel table summarizing a crop insurance demand system. The table is organized into panels for coverage level (Theta), insured acres (Gamma), total protection response, a covariance matrix block, and additional statistics. Coefficients are formatted as `estimate (std. error)` with significance stars.

Usage

```
format_fcip_demand_table(df, var_labels)
```

Arguments

df	A data frame containing the results with columns: • demand (chr): panel identifier; expected values include "Theta", "Gamma", and "Total". • coef (chr): raw coefficient/row labels (e.g., "tilda_rate", "residCov_11", "N"). • Estimate (dbl): point estimates. • StdError (dbl): standard errors (may be NA for scalars). • Pvalue (dbl): p-values used to add significance stars.
var_labels	A named character vector mapping raw names to display labels,

Details

Designed for README/output knitted as `github_document`; use with `knitr::kable(..., format = "pipe")` to avoid HTML-only features.

Value

A tibble with two columns, `Variables` and `Estimates`, where panel headers have empty `Estimates` to enable bolding (if rendered in HTML) and coefficients are formatted as `"estimate*** (se)"`.

get_yu2018_instrument *Formulate & Merge National Subsidy Rate Instrument (Yu et al., 2018)*

Description

Downloads the historical Summary of Business RDS and computes national subsidy-rate instruments at specified coverage levels, following Yu et al. (2018).

Usage

```
get_yu2018_instrument(
  dt,
  delivery_systems = c("RBUP", "FBUP"),
  plan_codes = c(1:3, 90, 44, 25, 42),
  coverage_levels = c(65, 75)
)
```

Arguments

dt	sobcov
delivery_systems	Character vector. Delivery systems to include; default `c("RBUP","FBUP")`.
plan_codes	Integer vector. Insurance plan codes to include; default `c(1:3, 90, 44, 25, 42)`.
coverage_levels	Numeric vector. Percent coverage levels to keep; default `c(65, 75)`.

Value

A data.table with columns: commodity_year, subsidy_rate_65, subsidy_rate_75.

See Also

Other FCIP instruments: `estimate_fcip_instruments()`

`global_variables` *fsa_crop_linker*

Description

A combined dataset for global_variables

Usage

```
global_variables
```

Format

A data frame with 8594 rows and 8 columns covering Inf–Inf.

Source

Internal innovation

nassAgPriceMonthlyIndex *nassAgPriceMonthlyIndex*

Description

A combined dataset for nassAgPriceMonthlyIndex

Usage

```
data(nassAgPriceMonthlyIndex)
```

Format

A data frame with 2645 rows and 8 columns covering NA-NA.

Source

USDA NASS: https://www.nass.usda.gov/Charts_and_Maps/graphics/data

nassSeasonLengthState *nassSeasonLengthState*

Description

A combined dataset for nassSeasonLengthState

Usage

```
data(nassSeasonLengthState)
```

Format

A data frame with 9523 rows and 14 columns covering 1981-2025.

Source

Calculated

nassSurveyInputApplications *nassSurveyInputApplications*

Description

A combined dataset for nassSurveyInputApplications

Usage

```
data(nassSurveyInputApplications)
```

Format

A data frame with 3830 rows and 25 columns covering 1990-2024.

Source

USDA NASS Quick Stats

nassSurveyMYAprice *nassSurveyMYAprice*

Description

A combined dataset for nassSurveyMYAprice

Usage

```
data(nassSurveyMYAprice)
```

Format

A data frame with 33384 rows and 7 columns covering 1866-2025.

Source

USDA NASS Quick Stats

`nassSurveyPriceRecivedIndex` *nassSurveyPriceRecivedIndex*

Description

A combined dataset for nassSurveyPriceRecivedIndex

Usage

```
data(nassSurveyPriceRecivedIndex)
```

Format

A data frame with 36 rows and 3 columns covering 1990-2025.

Source

USDA NASS Quick Stats

`nassSurveyRentalRates` *nassSurveyRentalRates*

Description

A combined dataset for nassSurveyRentalRates

Usage

```
data(nassSurveyRentalRates)
```

Format

A data frame with 1792 rows and 5 columns covering 1994-2025.

Source

Output from get_state_rental_rates() function

premium_subsidy_schedule *premium_subsidy_schedule*

Description

A combined dataset for premium_subsidy_schedule

Usage

```
data(premium_subsidy_schedule)
```

Format

A data frame with 8071 rows and 7 columns covering 2001-2027.

Source

USDA-RMA, Actuarial Data Master supplemented data from legacy ADM files

prevented_planting_adm *prevented_planting_adm*

Description

A combined dataset for prevented_planting_adm

Usage

```
data(prevented_planting_adm)
```

Format

A data frame with 252607 rows and 10 columns covering 2011-2027.

Source

USDA-RMA, Actuarial Data Master supplemented data from legacy ADM files

`rfcipDemand_controls` *Demand Model Control Parameters*

Description

Constructs a control list for demand-model routines in **rfcipDemand**, primarily governing admissible ranges for elasticity parameters used in demand adjustments and simulations.

If `elasticity_limits` is not supplied, the function defaults to `fcip_elasticity_limits`, a package-level object defining program-consistent elasticity bounds.

Usage

```
rfcipDemand_controls(elasticity_limits = NULL)
```

Arguments

`elasticity_limits`
: Optional named list specifying lower and upper bounds for elasticity parameters. If `NULL`, defaults to `fcip_elasticity_limits`.

Value

A named list of control parameters, ready to be passed to other simulation functions.

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